

General Relativity

Final Examination Spring 2021 Total Possible Score: 115

Name: _____ Class: _____ Student number: _____

1. (25 points)

- (a) Consider spacetime with curvature due only to electromagnetic radiation. The energy-momentum tensor for electromagnetism is given by

$$T_{\mu\nu} = F_{\mu\rho} F_{\nu}^{\rho} - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma} F^{\rho\sigma}$$

SHOW carefully that the Ricci scalar is zero.

Hint: begin by contracting the indices in Einstein's field Equation, Eqn. (4.44).

- (b) For dust, a perfect fluid, show that energy-momentum 'conservation' implies that the dust's four-velocity obeys the geodesic equation.
- (c) If the dust satisfies the Strong Energy Condition (SEC), what condition can you put on the Ricci scalar?
- (d) For a more general perfect fluid with non-zero pressure P satisfying SEC, what condition can you put on the Ricci scalar?

2. (15 points)

- (a) Demonstrate that any two-dimensional spacetime manifold is maximally symmetric by showing that the terms of the rank-(0,4) Riemann curvature tensor give exactly one non-zero magnitude. Thus any two-dimensional manifold, such as for instance the surface of a sphere, is automatically maximally-symmetric.
- (b) Given this, show that any two-dimensional spacetime 'has no dynamics' – that is, it has no matter-energy content.

3. (40 points) Consider the 2-dimensional line element $ds^2 = a^2 \left(\frac{dx^2}{1+x^2} + x^2 dy^2 \right)$, where a is a constant.

- (a) Write down the metric and the inverse metric.
- (b) Find all Christoffel symbols and the geodesic equations.
- (c) Find a non-zero Riemann curvature tensor term.
- (d) Find the rank-(0,4) tensor corresponding to your answer in part (c).
- (e) Find all Ricci tensor terms and also the Ricci scalar.
- (f) Show that the vector $V^i \equiv (V^x, V^y) = (0, 1)$ is a Killing vector.
- (g) For what values of the constant K is the vector

$$W^i \equiv (W^x, W^y) = \left(K\sqrt{1+x^2} \cos y, \frac{\sqrt{1+x^2}}{x} \sin y \right)$$

a Killing vector for this metric?

4. (35 points) Consider again the line element given by

$$ds^2 = e^{-2\zeta} dt^2 - e^{+2\zeta} (dx^2 + dy^2 + dz^2)$$

as on the pre-final exam, except this time the parameter ζ is time-dependent, $\zeta = \zeta(t)$; ζ has **no** spatial dependence.

- (a) Find the Christoffel symbols and the geodesic equations.
- (b) Find **every** non-zero term of the Riemann curvature tensor and also find all the corresponding rank-(0,4) tensors.
- (c) Find all components of the Ricci tensor.
- (d) Find the Ricci scalar.
- (e) Find all components of the Einstein tensor.